

Ultima 2.0
Cooperative OS
Simulator

Scheduler/Semaphore
Phase 1

Implements task management and synchronization using a round-robin scheduler and semaphores to control access to shared resource

Message Passing
Phase 2

Implements inter-process communication through task mailboxes and message queues, allowing tasks to send and receive structured messages for safe communication and coordination during execution

Memory Management
Phase 3

Implements memory management using an MMU, allowing tasks to request, use, and release memory. Uses a first-fit allocation strategy and tracks memory with linked lists, while semaphores ensure safe access during concurrent execution

File Systems
Phase 4